

ASSIGNMENT-10.5

Lab 10 – Code Review and Quality: Using AI to Improve Code

Quality and Readability

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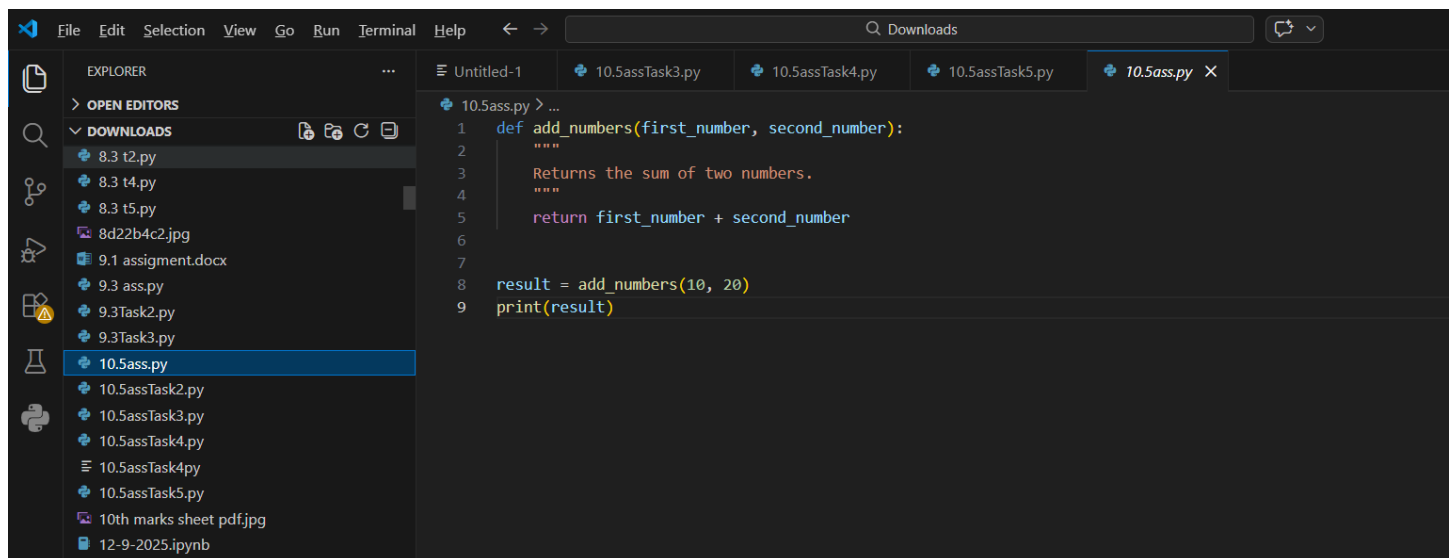
Task 1: Improving Variable & Function Names Original Code

```
def f(a, b):    return
```

```
a + b print(f(10,
```

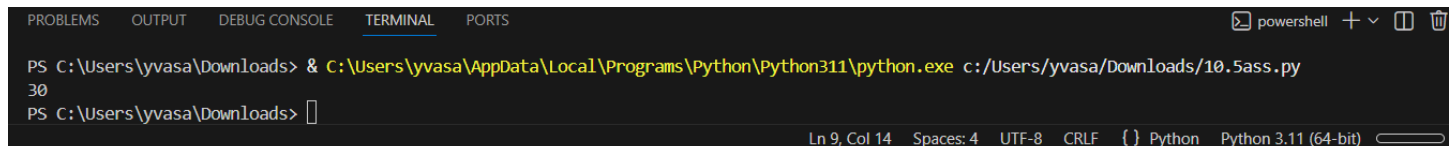
```
20))
```

CODE:



```
1 def add_numbers(first_number, second_number):
2     """
3     Returns the sum of two numbers.
4     """
5     return first_number + second_number
6
7
8 result = add_numbers(10, 20)
9 print(result)
```

OUTPUT



```
PS C:\Users\yvasa\Downloads> & C:\Users\yvasa\AppData\Local\Programs\Python\Python311\python.exe c:/Users/yvasa/Downloads/10.5ass.py
30
PS C:\Users\yvasa\Downloads>
```

Task Description #2 – Missing Error Handling

Task: Use AI to add proper error handling.

Sample Input Code: def

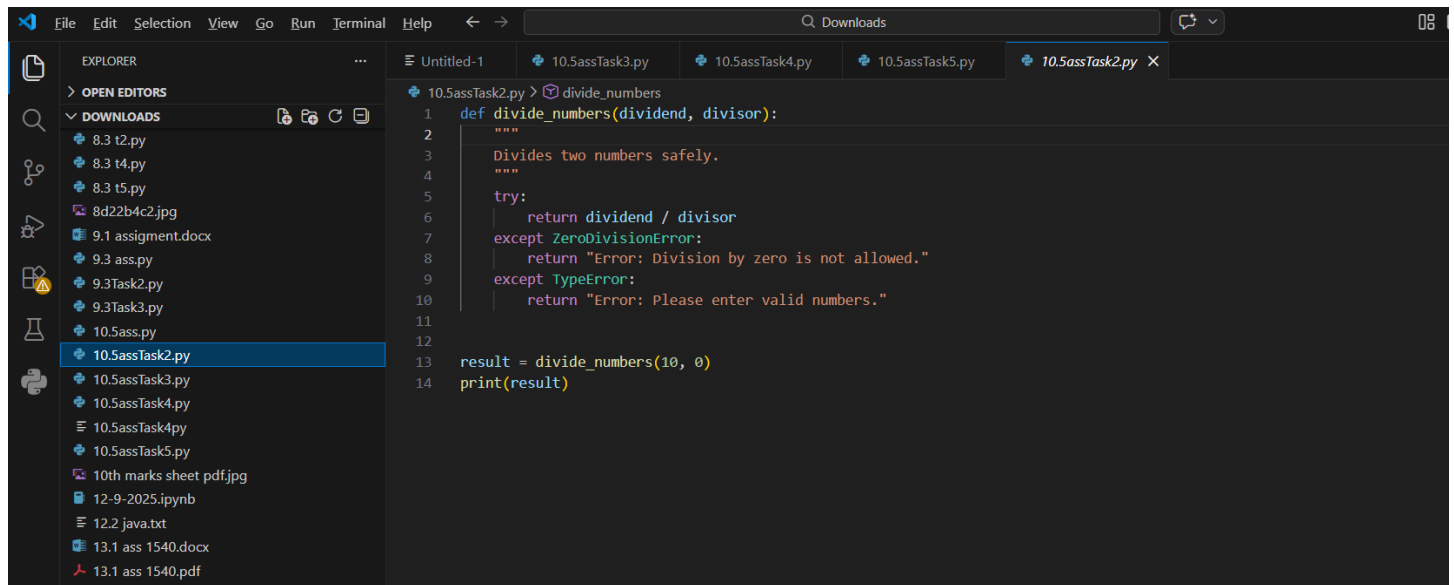
divide(a, b): return a / b

print(divide(10, 0))

Expected Output:

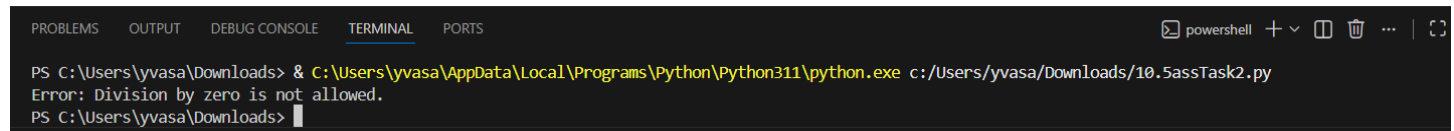
- Code with exception handling and clear error messages

CODE :



```
1 def divide_numbers(dividend, divisor):
2     """
3     Divides two numbers safely.
4     """
5
6     try:
7         return dividend / divisor
8     except ZeroDivisionError:
9         return "Error: Division by zero is not allowed."
10    except TypeError:
11        return "Error: Please enter valid numbers."
12
13    result = divide_numbers(10, 0)
14    print(result)
```

OUTPUT:



```
PS C:\Users\yvasa\Downloads> & C:\Users\yvasa\AppData\Local\Programs\Python\Python311\python.exe c:/Users/yvasa/Downloads/10.5assTask2.py
Error: Division by zero is not allowed.
PS C:\Users\yvasa\Downloads>
```

Task Description #3: Student Marks Processing System

The following program calculates total, average, and grade of a student, but it has poor readability, style issues, and no error handling.

marks=[78,85,90,66,88] t=0

for i in marks:

t=t+i

a=t/len(marks) if

a>=90: print("A")

elif a>=75:

```
print("B") elif
```

```
a>=60: print("C")
```

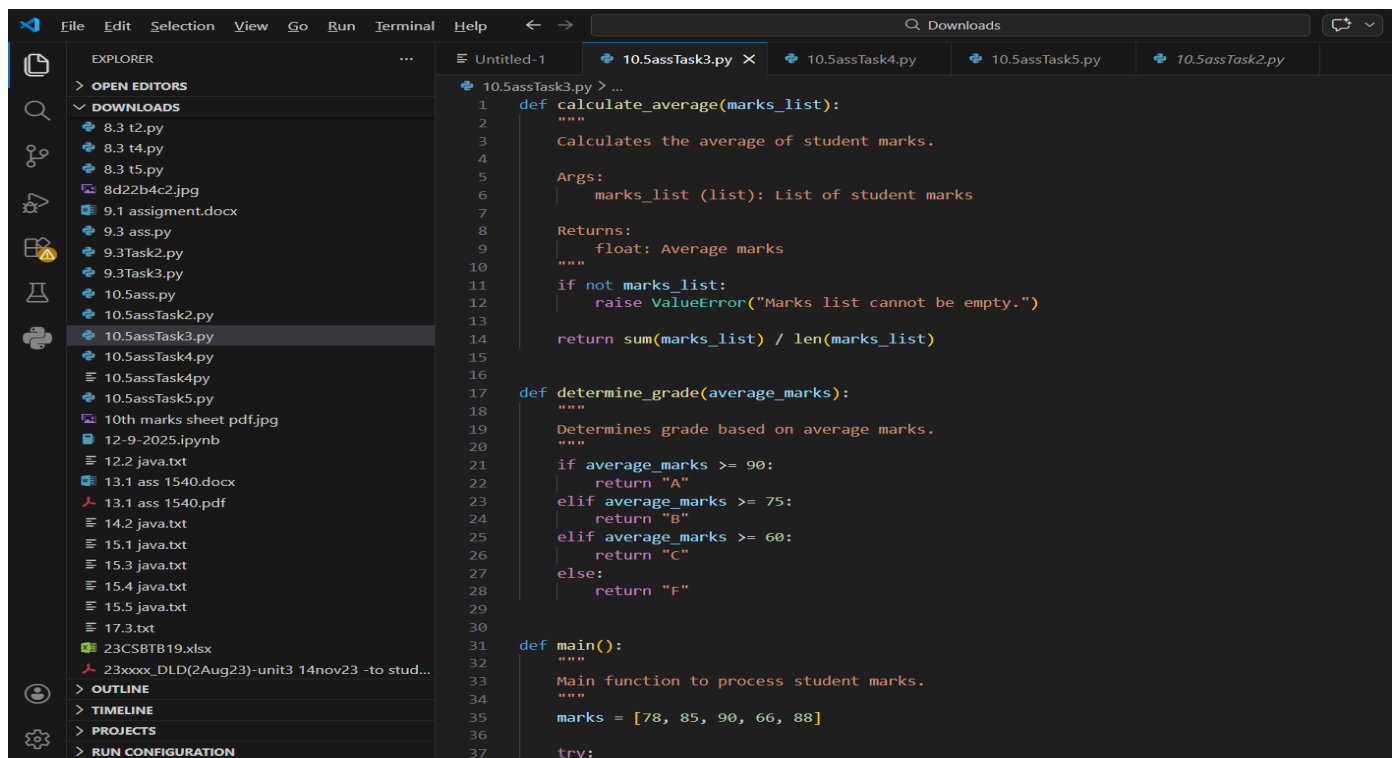
```
else:
```

```
print("F")
```

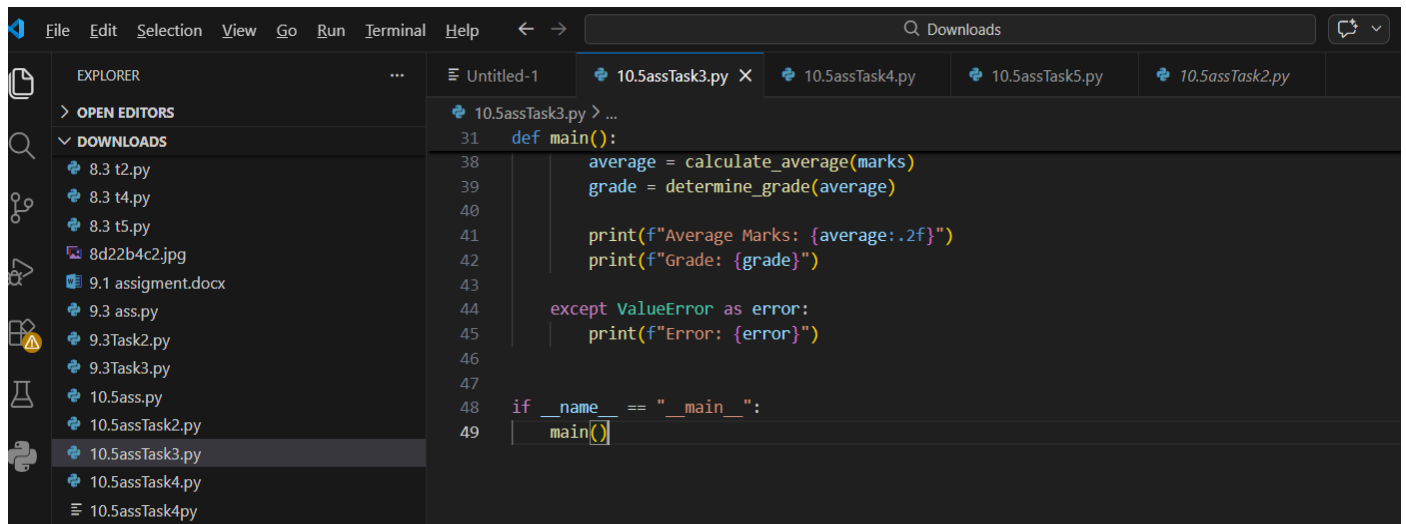
Task:

- Use AI to refactor the code to follow PEP 8 standards.
- Add meaningful variable names, functions, and comments.
- Add basic input validation and documentation.

CODE :



```
1 def calculate_average(marks_list):
2     """
3     Calculates the average of student marks.
4
5     Args:
6     | marks_list (list): List of student marks
7
8     Returns:
9     | float: Average marks
10    """
11    if not marks_list:
12        raise ValueError("Marks list cannot be empty.")
13
14    return sum(marks_list) / len(marks_list)
15
16
17 def determine_grade(average_marks):
18     """
19     Determines grade based on average marks.
20
21     if average_marks >= 90:
22         return "A"
23     elif average_marks >= 75:
24         return "B"
25     elif average_marks >= 60:
26         return "C"
27     else:
28         return "F"
29
30
31 def main():
32     """
33     Main function to process student marks.
34     """
35     marks = [78, 85, 90, 66, 88]
36
37     try:
```



```
31 def main():
32     """
33     Main function to process student marks.
34     """
35     marks = [78, 85, 90, 66, 88]
36
37     try:
38         average = calculate_average(marks)
39         grade = determine_grade(average)
40
41         print(f"Average Marks: {average:.2f}")
42         print(f"Grade: {grade}")
43
44     except ValueError as error:
45         print(f"Error: {error}")
46
47
48 if __name__ == "__main__":
49     main()
```

OUTPUT:

```
PS C:\Users\yvasa\Downloads> & C:\Users\yvasa\AppData\Local\Programs\Python\Python311\python.exe c:/Users/yvasa/Downloads/10.5assTask3.py
Average Marks: 81.40
Grade: B
PS C:\Users\yvasa\Downloads> |
```

Task Description #4: Use AI to add docstrings and inline comments to the

following function. def factorial(n):

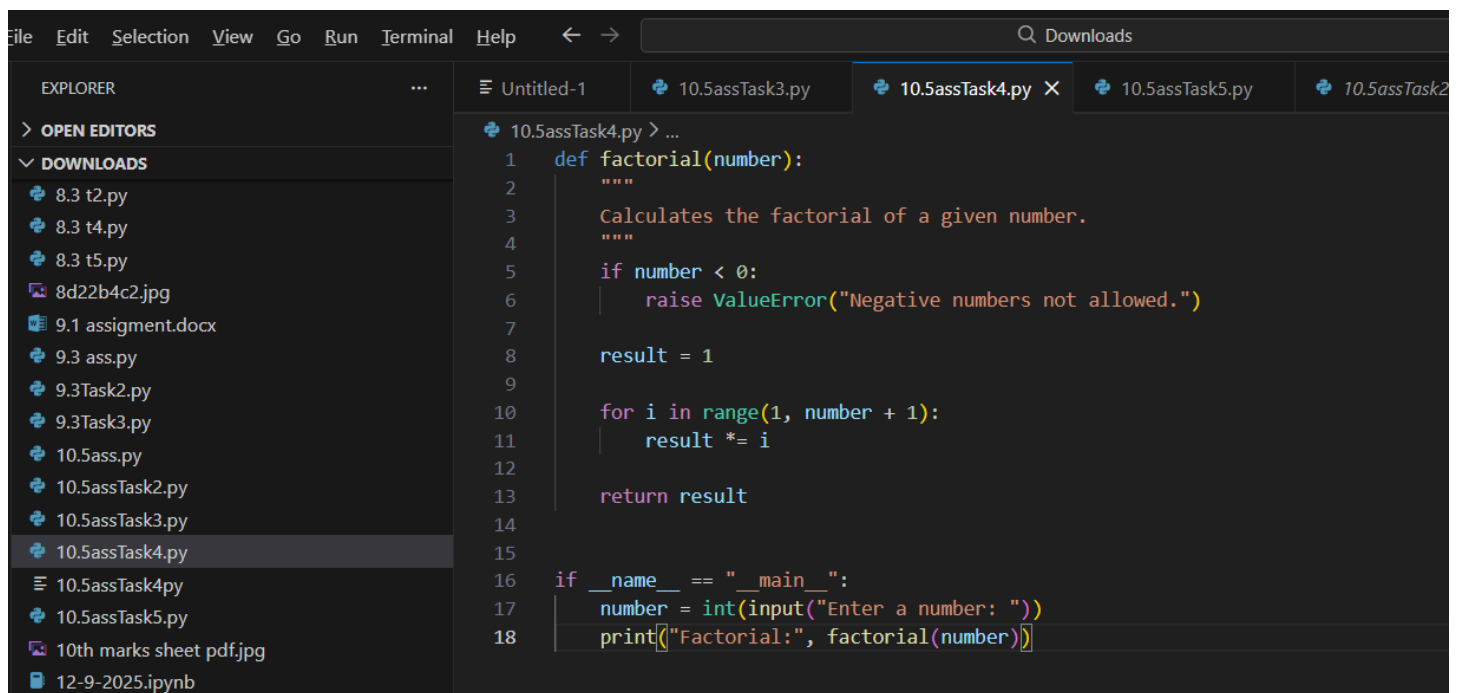
result = 1 for i in

range(1,n+1):

result *= i

return result

CODE :



The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a list of files in the 'Downloads' folder, including '10.5assTask4.py'. The code editor shows the following Python code:

```
1 def factorial(number):
2     """
3     Calculates the factorial of a given number.
4     """
5     if number < 0:
6         raise ValueError("Negative numbers not allowed.")
7
8     result = 1
9
10    for i in range(1, number + 1):
11        result *= i
12
13    return result
14
15
16 if __name__ == "__main__":
17     number = int(input("Enter a number: "))
18     print(f"Factorial:", factorial(number))
```

OUTPUT:

```
PS C:\Users\yvasa\Downloads> & C:\Users\yvasa\AppData\Local\Programs\Python\Python311\python.exe c:/Users/yvasa/Downloads/10.5assTask4.py
Enter a number: 5
Factorial: 120
PS C:\Users\yvasa\Downloads> |
```

Task Description #5: Password Validation System (Enhanced)

The following

Python program validates a password using only a minimum length check,

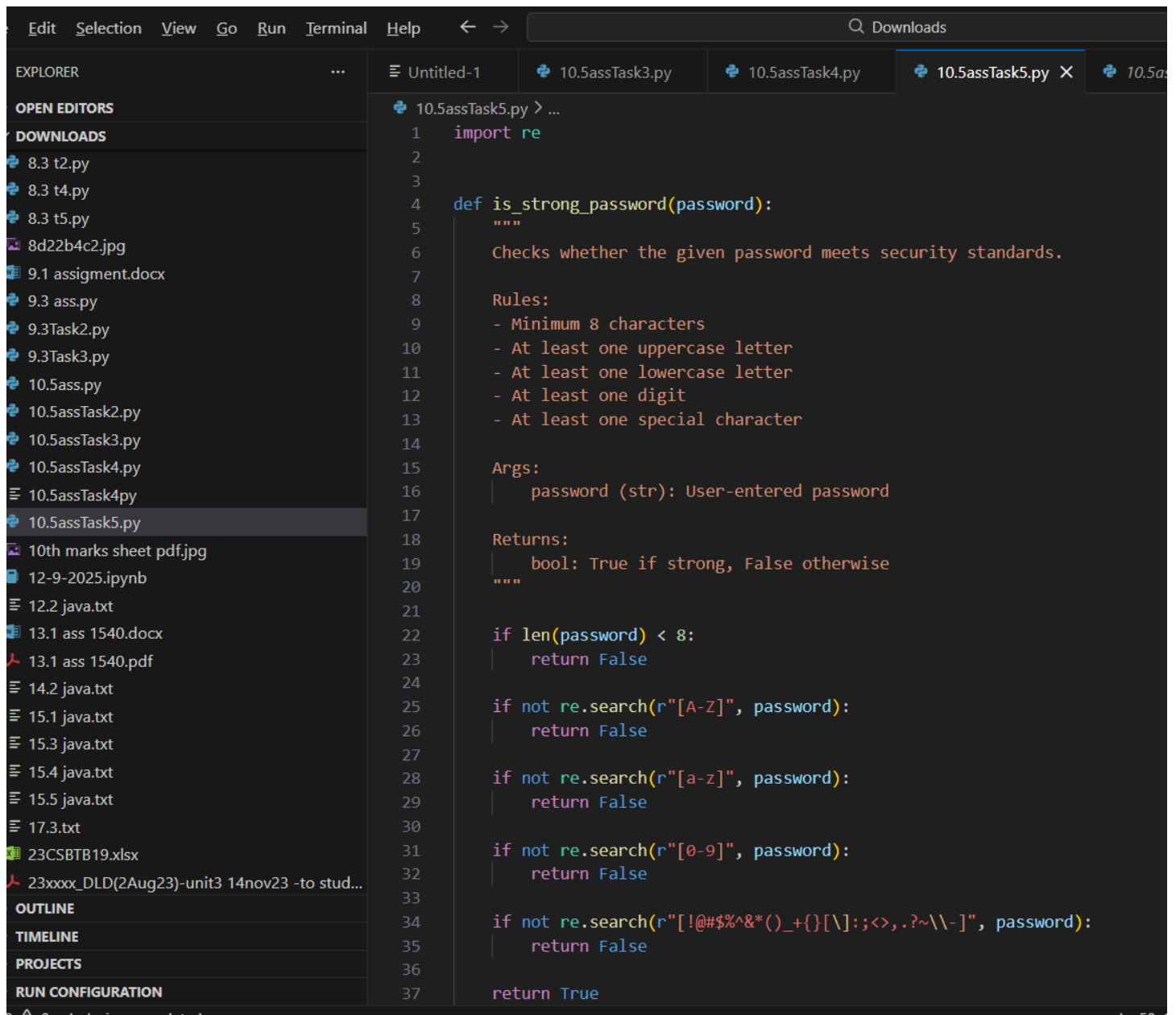
which is insufficient for real-world security requirements.

```
pwd = input("Enter password: ") if
```

```
len(pwd) >= 8: print("Strong") else:
```

```
print("Weak")
```

CODE :



The screenshot shows a code editor with a dark theme. The Explorer panel on the left lists various files, including Python scripts and documents. The main editor area displays the code for '10.5assTask5.py'. The code defines a function 'is_strong_password' that checks password security based on length, uppercase letters, lowercase letters, digits, and special characters. The code is as follows:

```
1 import re
2
3
4 def is_strong_password(password):
5     """
6     Checks whether the given password meets security standards.
7
8     Rules:
9     - Minimum 8 characters
10    - At least one uppercase letter
11    - At least one lowercase letter
12    - At least one digit
13    - At least one special character
14
15    Args:
16    | password (str): User-entered password
17
18    Returns:
19    | bool: True if strong, False otherwise
20    """
21
22    if len(password) < 8:
23        return False
24
25    if not re.search(r"[A-Z]", password):
26        return False
27
28    if not re.search(r"[a-z]", password):
29        return False
30
31    if not re.search(r"[0-9]", password):
32        return False
33
34    if not re.search(r"[!@#$$%^&*()_+{}[\]:;<>.,?~\\-]", password):
35        return False
36
37    return True
```

The screenshot shows a code editor with a dark theme. The top menu bar includes 'Selection', 'View', 'Go', 'Run', 'Terminal', and 'Help'. Below the menu is a toolbar with a search icon and the text 'Downloads'. The editor has several tabs open: 'Untitled-1', '10.5assTask3.py', '10.5assTask4.py', '10.5assTask5.py' (which is the active tab), and '10.5assTa...'. The left sidebar shows a file explorer with various files, including '10.5assTask5.py' which is selected. The main editor area displays the following Python code:

```
40 def main():
41     """
42     Main function for password validation.
43     """
44     user_password = input("Enter password: ")
45
46     if is_strong_password(user_password):
47         print("Password is Strong ✅")
48     else:
49         print("Password is Weak ❌")
50         print("It must contain:")
51         print("- At least 8 characters")
52         print("- One uppercase letter")
53         print("- One lowercase letter")
54         print("- One digit")
55         print("- One special character")
56
57
58 if __name__ == "__main__":
59     main()
```

OUTPUT

The screenshot shows a terminal window with a dark theme. The top bar includes 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is the active tab), and 'PORTS'. The terminal shows the following output:

```
Factorial: 120
PS C:\Users\yvasa\Downloads> & C:\Users\yvasa\AppData\Local\Programs\Python\Python311\python.exe c:/Users/yvasa/Downloads/10.5assTask5.py
Enter password: sara123
Password is Weak ❌
It must contain:
- At least 8 characters
- One uppercase letter
- One lowercase letter
- One digit
- One special character
PS C:\Users\yvasa\Downloads> & C:\Users\yvasa\AppData\Local\Programs\Python\Python311\python.exe c:/Users/yvasa/Downloads/10.5assTask5.py
Enter password: Sahasra@143
Password is Strong ✅
PS C:\Users\yvasa\Downloads> |
```